MTH314

Timed Online Assignment – January Semester 2021

Real Analysis II

Tuesday, 25 May 2021 **1:00 pm – 3:30pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

(a) Give an example of a function $f: \mathbb{R} \longrightarrow \mathbb{R}$ such that for all integers n, $f(x)$ is differentiable on the open interval $(n, n+1)$ but not continuous at the point $x = n$.

(4 marks)

(b) Let $g(x) = \frac{1}{2x-3}$. Compute the value of $g'(0)$ from first principles, that is, by using the limit definition of the derivative.

(6 marks)

(c) Let $h(x) = x^3 - 4x^2 + 3/x$. Given that h is a strictly increasing function in an open interval containing the point $x = 3$ and that $h(3) = -8$, calculate the value of $(h^{-1})'(-8)$.

(5 marks)

Question 2

Let n, m be positive integers satisfying $\left(4\sqrt{2}\right)m < n < 6m$. Define a function

$$f(x) = mx + \frac{n}{\sqrt{x}}\,, \qquad x > 0\,.$$

(a) Find the interval on which f is increasing and the interval on which f is decreasing. Express your answers in terms of n and m.

(6 marks)

(b) Prove that the minimum value of $f(x)$ in the interval $[1,4]$ is given by

$$2^{-\frac{2}{3}}(3)m^{1/3}n^{2/3}$$

and also find the maximum value of $f(x)$ in the interval $[1,4]$.

(7 marks)

(c) Show that

$$|f(x) - (m+n)| \geq \frac{\left(n - \left(4\sqrt{2}\right)m\right)}{\pi}\left(\sin\left(\frac{\pi x}{4}\right) - \frac{1}{\sqrt{2}}\right)$$

whenever $1 < x < 2$.

(7 marks)

Question 3

Let n be a positive integer. Suppose that $f: \left[\frac{\pi}{4}, \frac{5\pi}{12}\right] \longrightarrow \mathbb{R}$ is a function that is differentiable 3 times on the closed interval $\left[\frac{\pi}{4}, \frac{5\pi}{12}\right]$, and which satisfies $f(\pi/3) = \sqrt{3}$, $f'(\pi/3) = 4$, $f''(\pi/3) = 8\sqrt{3}$, and $|f'''(x)| \leq n$ for $\pi/4 \leq x \leq 5\pi/12$.

(a) Find the Taylor polynomial $P_2(x)$ of $f(x)$ about the point $x = \pi/3$, that is, the Taylor expansion about $\pi/3$ up to and including the term involving $(x - \pi/3)^2$.

(3 marks)

(b) Find the maximum error in estimating $f(x)$ in the domain $\pi/4 \leq x \leq 5\pi/12$ using the Taylor polynomial $P_2(x)$ about the point $x = \pi/3$. Express your answer in exact form in terms of n.

(3 marks)

(c) Use the Taylor polynomial $P_2(x)$ about the point $x = \pi/3$ to estimate the value of $f(1)$, giving your answer to 2 decimal places.

(2 marks)

(d) The function $g(x) = \tan(x)$ is infinitely differentiable on $\left[\frac{\pi}{4}, \frac{5\pi}{12}\right]$, and satisfies $g(\pi/3) = \sqrt{3}$, $g'(\pi/3) = 4$, $g''(\pi/3) = 8\sqrt{3}$. Determine the smallest possible value of the integer n such that $|g'''(x)| \leq n$ for $\pi/4 \leq x \leq 5\pi/12$, and determine whether the estimate obtained in part (c) provides an approximate value of $g(1)$ that is accurate to 2 decimal places.

(7 marks)

Question 4

(a) Compute the limit

$$\lim_{x \to 0^+} \left(\frac{\ln(x+1) - \sin(x)}{x^2} \right)$$

using L'Hospital's Rule. Full working is required.

(7 marks)

(b) Compute the limit

$$\lim_{n \to \infty} \left[\frac{1}{n} \sum_{i=1}^{n} \ln\left(\frac{i}{n} + 2\right) \right],$$

expressing your answer in exact form. Full working is required.

(8 marks)

Question 5

Define a function

$$f(x) = 5|x| + x - x^2\,, \qquad -4 \le x \le 6\,.$$

A graph of the function $y = f(x)$ is provided in Figure Q5 below. The graph $y = f(x)$ intersects the x-axis at the points $(-4,0)$, $(0,0)$ and $(6,0)$.

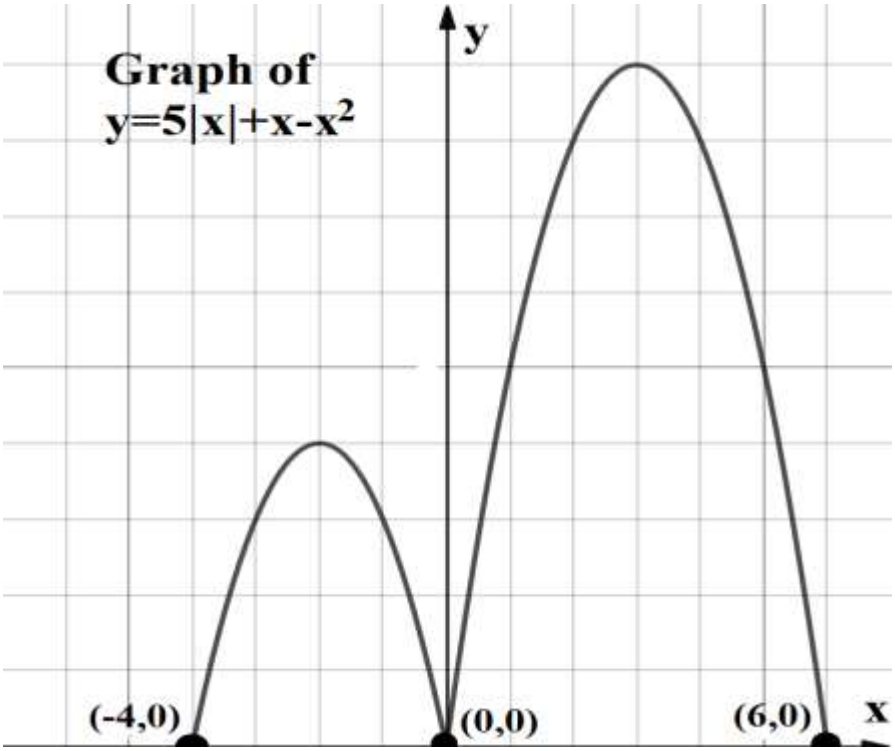

Figure Q5

(a) Let $\mathcal{P}$ denote the tagged partition of $[-4,6]$ into 6 intervals, consisting of the intervals and tags given below:
$([-4,-3]$, tag: $-3.5)$, $([-3,-1.5]$, tag: $-2)$, $([-1.5,0]$, tag: $-1)$,
$([0,2]$, tag: $1)$, $([2,4.5]$, tag: $3)$, $([4.5,6]$, tag: $5.5)$.
Calculate the value of the Riemann sum $\mathcal{S}(f,\mathcal{P})$. Full working is required.

(6 marks)

(b) Let Q denote the partition of $[-4,6]$ into 5 equal-width intervals, namely,

$$[-4,-2], [-2,0], [0,2], [2,4], [4,6]\,.$$

Calculate the value of the Lower Riemann sum $L(f,Q)$ and the Upper Riemann sum $U(f,Q)$. Full working is required.

(6 marks)

(c) Find a differentiable function $g\colon [-4,6] \longrightarrow \mathbb{R}$ such that $g(0) = 1$ and $g'(x) = f(x)$ for all $x \in [-4,6]$.

(8 marks)

Question 6

Let n be an integer greater than 1, and let a, b be positive real numbers with $0 < a < b$. Define a function $f: [a, b] \longrightarrow \mathbb{R}$, $f(x) = x^n$. For any positive integer m and for any choice of points $x_0, x_1, x_2, \dots, x_m$ satisfying $a = x_0 < x_1 < x_2 < \cdots < x_m = b$, define a tagged partition $\mathcal{P}_m(x_0, x_1, x_2, \dots, x_m) = \{ ([x_{i-1}, x_i], t_i) : 1 \leq i \leq m \}$ of $[a, b]$, where

$$t_i = \left[\frac{1}{n+1}((x_{i-1})^n + (x_{i-1})^{n-1}(x_i)^1 + (x_{i-1})^{n-2}(x_i)^2 + \cdots + (x_i)^n)\right]^{1/n}$$

for each integer i, $1 \leq i \leq m$.

(a) For each tagged partition $\mathcal{P}_m(x_0, x_1, x_2, \dots, x_m)$, provide a detailed verification that $x_{i-1} < t_i < x_i$ for all i, $1 \leq i \leq m$.

(6 marks)

(b) It is known that for any positive real number ε, there exists a positive real number δ such that for any tagged partition Q of $[a, b]$ with mesh $\|Q\| < \delta$, there exists some positive integer m and a tagged partition $\mathcal{P}_m(x_0, x_1, x_2, \dots, x_m)$ such that

$$\left|\mathcal{S}\big(f, \mathcal{P}_m(x_0, x_1, x_2, \dots, x_m)\big) - \mathcal{S}(f, Q)\right| < \varepsilon.$$

Demonstrate from first principles using the above property that f is Riemann integrable on $[a, b]$, and that

$$\int_a^b f = \frac{1}{n+1}(b^{n+1} - a^{n+1}).$$

(9 marks)

----- END OF PAPER -----